

CONTROLLABLE ELECTRONIC LOAD CIRCUIT FOR DC POWER SUPPLY

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If you are interested in testing voltage-regulated power supplies under loaded conditions, here is a simple electronic load circuit controlla-

on a large-area heat-sink, can be used for testing any DC power supply in the range of 2 to 50 volts capable of delivering up to 10A current. You can increase the current limit to 20 amps by adding two more 2N3773

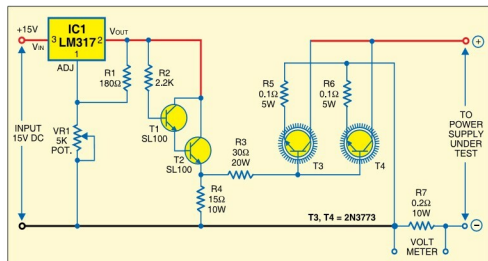


Fig. 1: Electronic load circuit controllable from a single supply

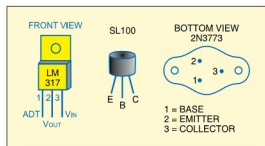


Fig. 2: Pin configurations of LM317, SL100 and power transistor 2N3773

ble from a single 2-12V, 200mA power supply. The variable power supply is generated from regulator LM317.

This electronic load circuit, using a parallel combination of power transistors T3 and T4 (2N3773) mounted

power transistors (not shown in the circuit) in parallel, mounted on a suitable heat-sink, without making any change in the controllable drive circuit.

This test circuit has two main advantages: elimination of high-wattage resistors normally required for the load and controllability of the load current.

As shown in Fig. 1, transistors T1 and T2 form a Darlington pair which is powered by variable 12V using a three-terminal regulator LM317. Resistor R2 provides base current to the Darlington pair drive circuit, which is controlled by 5-kilo-ohm, 10-turn

potentiometer VR1 connected to the ADJ terminal of LM317. The output of the driver circuit is taken across resistor R4 and fed to power transistors T3 and T4 through current-limiting resistor R3.

The power supply under test is connected across collector and emitter leads of the power transistors as shown in Fig. 1. Load current is sensed by small wire-wound resistor R7 (0.2-ohm, 10W) connected in series with the load and the voltage drop across it is measured by a digital voltmeter. Thus a load current of 1 amp would give a drop of 100 mV, and so on. Thus a voltmeter reading of 1 mV means that current flowing in the power supply under test is 10 mA.

Assemble the control circuit on a general-purpose PCB and enclose in a box. Fix the potmeter on front side of the box to control load of the DC power supply to be tested. Fix the power transistor along with a large heat-sink with proper cooling arrangement. Draw two leads across resistor R7 for measuring the voltage through digital voltmeter.

EFY note. The heat-sink has to be air-cooled and the power dissipation of each 2N3773 power transistor should not be allowed to exceed 120 watts, although its rated value is 150 watts. ●



EARTH LEAKAGE TESTER

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Earth leakage from electrical wiring is a serious problem, especially during rainy season.

Pipelines are more vulnerable to earth leakage and may cause an unexpected electric shock. Electrical appliances or faulty wiring may be the source of leakage current through the wet

wall to the earth. An ordinary AC tester cannot detect the earth leakage if

